

Computational fluid–structure interaction analysis of passive adaptive blades in turbomachinery applications

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Abstract This chapter presents three applications of computational fluid-structure interaction (CFSI) in the field of turbomachinery. We explore novel designs for morphing blades that adapt to changes in flow direction, focusing on small-size reversible fans and turbines. The model framework is based on the finite element formulations of fluid dynamics, structural mechanics, and mesh moving equations, while a block-iterative approach is used for the FSI coupling. We conduct first a 2D study of a reversible fan cascade made of low-stiffness material. The goal is to achieve a stable passive change in airfoil curvature in response to the aerodynamic forces. A similar design solution is then investigated for a Wells type turbine. A 2D cascade study verifies the feasibility of the concept and explores the use of different material layouts. Then, we test the 3D blade under time-dependent flow rate conditions, simulating operation of the turbine in an oscillating water column facility for sea wave energy conversion. In all cases, the results show that using CFSI provides useful insight into the functioning of these devices.

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1 Introduction

Turbomachinery rotors are typically designed to operate under stationary conditions, and off-design operation and transients are kept as short as possible. In these cases, blade geometries are often assumed to be rigid. However, some applications require the machine to operate under continuously varying conditions, such as renewable energy turbines (e.g. wind turbines or Wells turbines) and reversible axial fans. If the blade geometry is rigid in these cases, the machine will only operate at maximum efficiency for short periods of time. Therefore, variable blade geometries are preferred.

A famous example of variable geometry in turbomachinery is a rotor system with variable blade pitch. However, this system can be expensive and technologically complex to implement, making it unsuitable for small-scale machines where construction costs and complexity must be minimized. An interesting and cost-effective solution is to design blades with passive-adaptive geometry (smart-blades). The elastic deformation of the structure after a change in the aerodynamic field is designed so that they enhance the machine's performance compared to fixed geometry.

Although smart-blade technology is already a mature and widely used technology in fields as wind energy and aeronautics, the same cannot be said for small turbomachinery, where high speeds, small sizes, and low costs make the assessment and optimization of the FSI response a difficult challenge, which is rarely investigated.

In this chapter, we present a three examples of FSI simulation applied to small rotor machines with morphing blades. A first 2D study is dedicated to a reversible flow fan, while 2D and 3D simulations are carried out to investigate the use of passive morphing blades on Wells turbine.

Reversible fans are a well-established technology that are widely used for air supply and extraction [1, 2]. However, reversible fan blades are often characterized by large thickness at the trailing edge, which is required for symmetry and flow reversibility. This design choice results in a loss of at least 5% in total efficiency compared to the non-reversible version of the same airfoil [3]. Using a sufficiently thin and flexible blade, it may be possible to adapt the section curvature through the effect of flow pressure, recovering some of the performance losses in both direct and reverse inflow directions.

Wells turbines represent another consolidated technology, and are commonly employed for wave energy conversion in Oscillating Water Columns (OWC) systems. Traditional Wells turbines present limits in terms of average efficiency, because they also have rigid and symmetrical blades, working for most of the time in off-design flow conditions.

In both of the above-mentioned cases, the efficiency of the rotors is low, because of the requirement of having symmetric and untwisted blades that induce large flow separation for most of the operating flow-rates and a not-optimal lift to drag ratio of the blade sections. A non-symmetric but rigid blade profile could improve the performance of the machine only for specific flow-rates, but it would reduce the performance when the main flow reverts its direction. Using a flexible smart-blade made of plastic material can improve the blade aerodynamic efficiency for a large

range of flow rates in both inflow directions. Such blade could also improve the reliability of the machine in a highly corrosive environment such as saline aerosol, and it can reduce the capital investment and operative costs because of the adoption of cheaper materials.

Powerful methods developed for FSI and flow analysis have been applied to many classes of turbomachinery problems. The classes of problems include wind turbines [4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26], fans [27], turbochargers [28, 29, 30, 31], tidal-stream turbines [32], gas turbines [33], and pumps [34, 35]. The FSI of a passive morphing attachment applied to improve the performance of a large industrial ventilation fan has been investigated by the authors in [36].

In this study, the finite element method is applied to define all the discretized formulations for the three-field problem defined by solving the fluid dynamics, structure mechanics, and moving mesh equations. The aerodynamic subsystem is modelled using the Arbitrary Lagrangian–Eulerian (ALE) formulation of the Navier–Stokes equation obtained applying the Residual Based Variational Multiscale (RB-VMS) method [37, 38, 39], which subsumes the Streamline-Upwind/Petrov-Galerkin (SUPG) [40] and Pressure-Stabilizing/Petrov-Galerkin (PSPG) [41] stabilizations. The structure is modelled as a solid using the Total Lagrangian (TL) formulation of the elastic solid dynamics. The constitutive model implemented for this study is the Saint Venant–Kirchhoff model [10]. The mesh moving is solved using the mesh-Jacobian-based stiffening [42, 43, 44, 45]. Finally, we adopt a block-iterative method for the FSI coupling [46, 47, 48, 10].

The chapter is divided in a Section 2 reporting the governing equations for the three fields FSI problem, a Section 3 describing the adopted CFSI method and modelling, followed by Section 4 presenting the computations setup and results for: a 2D reversible fan cascade (Section 4.1), a 2D Wells turbine cascade (Section 4.2), and the 3D Wells turbine blade (Section 4.3). The last Section 5 reports a brief summary of the main outcomes of the study.

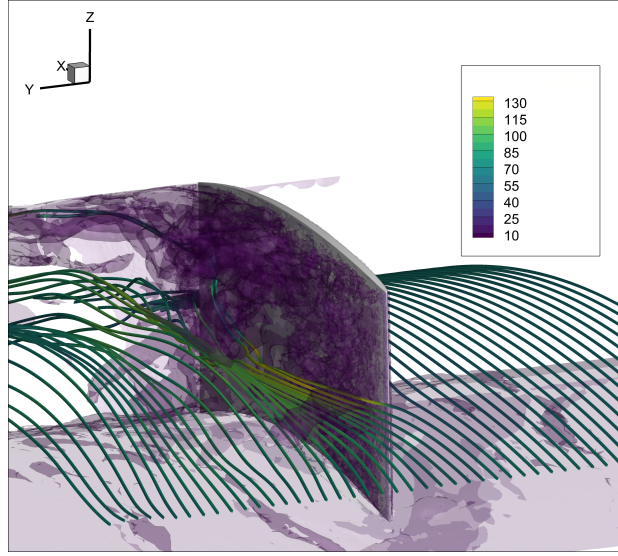


Fig. 16: Wells turbine 3D model. Flow patterns at t_1 . Isosurfaces of $\|\boldsymbol{\omega}\| = 0.1 \text{ s}^{-1}$ (violet) and the streamlines passing through the line $r = 0.22 \text{ m}$ of the periodic boundary, colored by the velocity magnitude (m/s).

Jacobian-based stiffening, and block-iterative FSI coupling. The weak enforcement of the Dirichlet boundary conditions is applied to handle time-dependent inflow velocity and no-slip boundary conditions in the fluid dynamics computations. The finite element formulation of structure mechanics is derived from the principle of virtual work applied to elastic solids, written with respect to the material domain of the structure in the reference configuration. The FSI coupling between the blocks of the fully discretized equation system representing the fluid mechanics, structural mechanics, and mesh moving equations is handled with the block-iterative coupling method, a method that is relatively easy to implement, but that performs efficiently for problems where the ratio between structure and fluid mass is large.

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